

3D picking for any industrial robot. A depth camera on the wrist and a small computer beside the robot find every part of the size you enter lying on a flat surface, number them in the pick order you choose, and bring the tool to the grip on each one. A connector puts this into the robot's own programs; the robot's program opens and closes the gripper.

Planning figures, not guarantees. The numbers below are deliberately conservative estimates derived from the software's limits, its programmed speeds and the camera's noise, except the ones marked ^M, measured on one bench cell. This release is a beta and nothing has been measured on a production cell. **Test with your parts, gripper and layout before you commit to them.**

COMPATIBLE ROBOTS

Arm	6-axis articulated	Cobot or industrial; grips top-down.
Reach	≥ 0.5 m^E	Rated; every grip is reach-checked.
Payload	+ 0.1 kg	Camera + bracket; CoG 40 mm off axis.
Tool flange	ISO 9409-1-50-4-M6 ISO 9409-1-80-6-M8	Brackets for both, under the gripper. Others need an adapter.
Controller	Ethernet, TCP	Its program opens a socket and moves.
Gripper	Two-finger	Any parallel jaw the program drives.

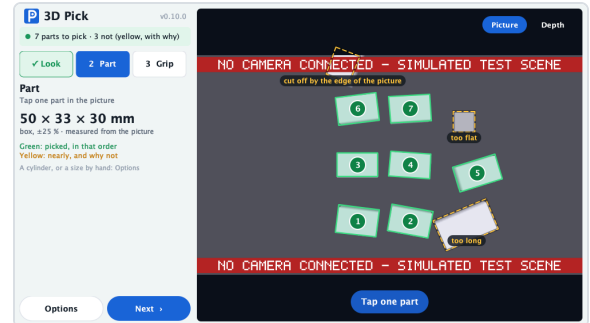
PICKING PERFORMANCE (by camera-to-part distance; verify on your cell)

	0.30 to 0.60 m working zone	1.0 to 1.5 m far limit	
Parts found	4 of 4^M	4 of 4^M	Colour + depth; 0 wrong. Depth alone: 1 of 4.
Size resolved to	±5 mm^E	±17 mm^E	It checks a size; it does not measure one.
Sizes told apart	≥ 10 mm^E	≥ 35 mm^E	Closer sizes read as the same part.
Smallest part found	20×20×8 mm^S	40×30×15^S	Every time; 60×40×20 to 1.2 m.
Lowest part	5 mm^E	15 mm^M	Above the surface; rough ones raise it.
Part centre	±4 mm^M	±10 mm^E	Across views; plus calibration: 9 mm per 1°.
Part heading	±2.5°^S	±8°^E	95 %. None for a square top.
Gap to neighbours	≥ 25 mm	≥ 40 mm^E	20 mm finger room a side.
Time to find	0.9 s^M	1.1 s^M	Per picture, Raspberry Pi 5.
Distance limits	0.28 to 1.6 m	No depth nearer; nothing farther is picked.	
Grip position	±10 mm^E	Plan the jaws for it. Mostly calibration.	
Part pose	Any face down	Boxes on any face; cylinders on end.	
Cycle, node only	15 s per part^E	To the grip. ~2 parts/min with a short place.	

^M Measured, a D435 on the camera computer: 50 × 30 × 30 mm foam on a UR3e, 0.32–0.50 m, straight down (read 49–51 × 29–31); boxes at 1.4 m. ^S Simulated: 720 parts, 12 cells per size and distance, a model of the D435's noise. ^E Estimate, not measured or guaranteed.

CAPABILITIES

- **Finds parts by volume**, depth with the colour outline: boxes on any face, cylinders on end; the floor is fitted round each part.
- **Any position, any rotation.** Fingers across the short side, or the long one.
- **Eight pick orders:** by rows or by columns, from any corner.
- **Checks before it moves:** reach by the arm's kinematics, finger room.
- **Says why not:** near misses drawn in yellow with the reason.
- **Queues what it saw:** no trip back to the picture point.
- **Runs inside the robot's own program** through its connector. No internet.



The Universal Robots connector on a PolyScope 5 pendant.

CONNECTORS

Universal Robots	PolyScope 5.4 to 5.26 · URCap 0.10.0 PolyScope X 10.8 to 10.14 · URCap 0.8.0
Other makes	On request

WHAT A CELL NEEDS

Camera	RealSense D435 (848 × 480, 30 fps, USB 3) on a printed bracket
Computer	Raspberry Pi class, beside the robot; ships set up
Network	Ethernet; TCP 7621 and 7622
Setup	One camera calibration; optional pick areas
Software	Free, open source (MIT). Kit from Advin.

NOT FOR (LIMITS OF THE DEPTH CAMERA)

- Measuring parts or quality checks
- Defect, surface or print inspection
- Telling apart near-equal sizes
- Thin or flat parts: sheets, washers, labels
- Transparent, mirror-like or very dark parts
- Black mats: not seen
- Translucent or shiny tops by depth alone
- Bin picking: piled or leaning parts
- Shapes other than boxes and upright cylinders
- Direct sunlight on the work area
- Driving the gripper or the place move

Download and details: perceptronics.advin.io · **Kit and questions:** nick@advin.io · **Source:** github.com/JimothyJohn/perceptronics

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